

Seminar Quiz 1

The Economy

Sciences Po Paris - Reims

Exercise 1

Table 1: Economy output

Agent	Production	Quantity	Price per Unit (\$)
Farmer	Trees	100	\$5
Lumberjack	Lumber (Domestic)	50	\$20
	Lumber (Imported)	25	\$25
Carpenter	Wooden Table (Domestic)	25	\$100
	Wooden Table (Exported)	5	\$130

Imagine an economy with three agents: a farmer, a lumberjack, and a carpenter. They each play a role in producing a final product, a wooden table.

The farmer grows trees in their plantation, which the lumberjack then produces lumber from. The carpenter uses the lumber to craft the wooden table. Additionally, the carpenter **imports** some of the lumber from abroad and **exports** some wooden tables. **Table 1** shows the output of this economy.

1. If the quality of the domestic and imported lumber are the same, what is the quantity of lumber required to produce a single wooden table? (1 point)

$$\frac{(50 + 25)}{(25 + 5)} = \frac{75}{30} \text{ or } 2.5$$

$$Y = C + G + I + NX \quad (1)$$

2. Calculate the GDP using the expenditure approach. Show the components of the formula correctly. (2 points)

$$Y = \underbrace{(25 * 100)}_C + \underbrace{(5 * 130)}_X - \underbrace{(25 * 25)}_M = \$2525$$

$$Y = \sum_{\text{for each producer}} VA \quad (2)$$

3. Calculate the GDP using the value-added approach. (2 points)

$$Y = VA_{\text{farmer}} + VA_{\text{lumberjack}} + VA_{\text{carpenter}} \text{ where,}$$

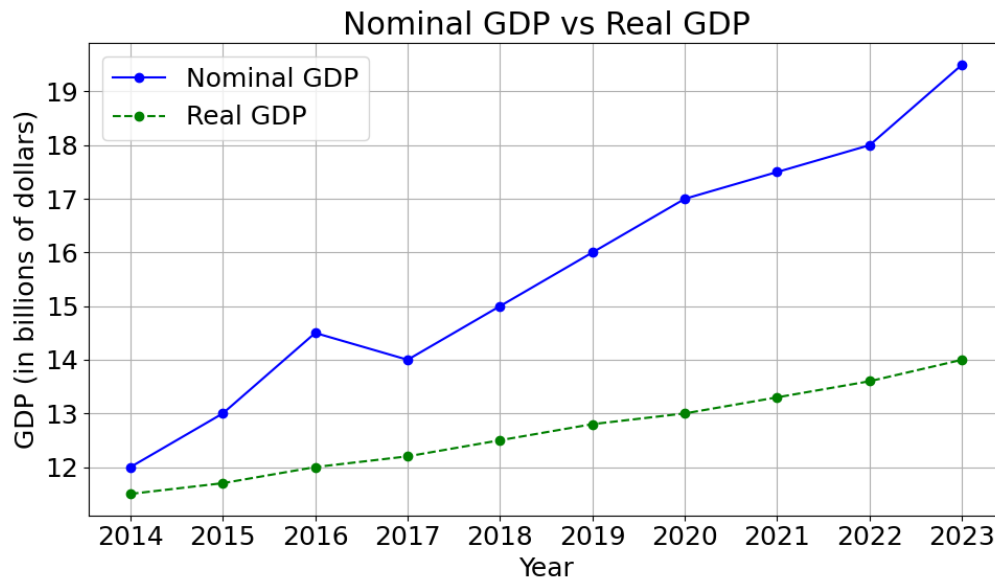
$$VA_{\text{farmer}} = 100 * 5 = \$500$$

$$VA_{\text{lumberjack}} = 50 * 20 - 500 = \$500$$

$$VA_{\text{carpenter}} = \underbrace{(25 * 100 + 5 * 130)}_{\text{price of final goods}} - \underbrace{(50 * 20 + 25 * 25)}_{\text{intermediate inputs}} = 3150 - (1000 + 625) = \$1525$$

therefore: $Y = 500 + 500 + 1525 = \$2525$

Exercise 2



Answer the questions below by looking at the graph above.

1. Which one of the following is most likely to be true for between 2014-2023? (2 points)

- A) The output of the economy has shrunk.
- B) One can buy more with \$100 in 2023, relative to 2014.
- C) One can buy less with \$100 in 2023, relative to 2014.
- D) The population has decreased.

2. Why do we distinguish between nominal and real GDP? What other information would we need to control for the demographic differences if we wanted to compare this economy with another? (3 points)

We distinguish between nominal/real GDP to account for the changes in prices (i.e. inflation/deflation). We would need population information to account for demographic differences (i.e. GDP per capita).